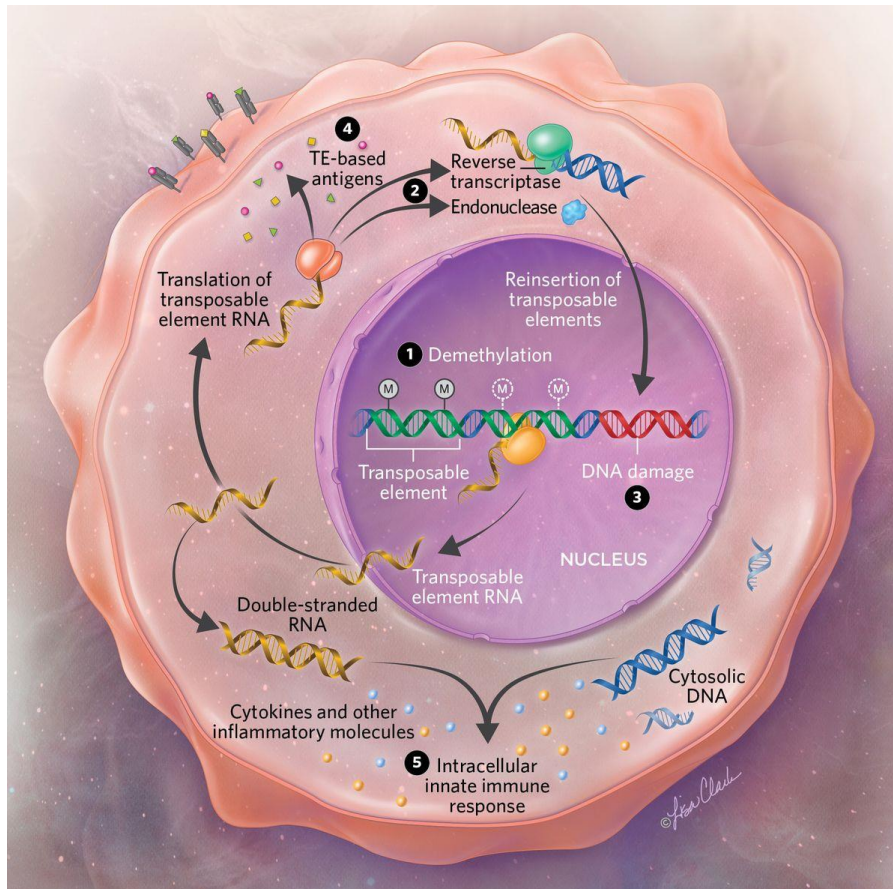


Transposable elements in human genetic disease

Transposable element insertions have been known to cause human genetic disease since the 1980s.

The scope of their contributions to heritable phenotypes is now starting to be uncovered.

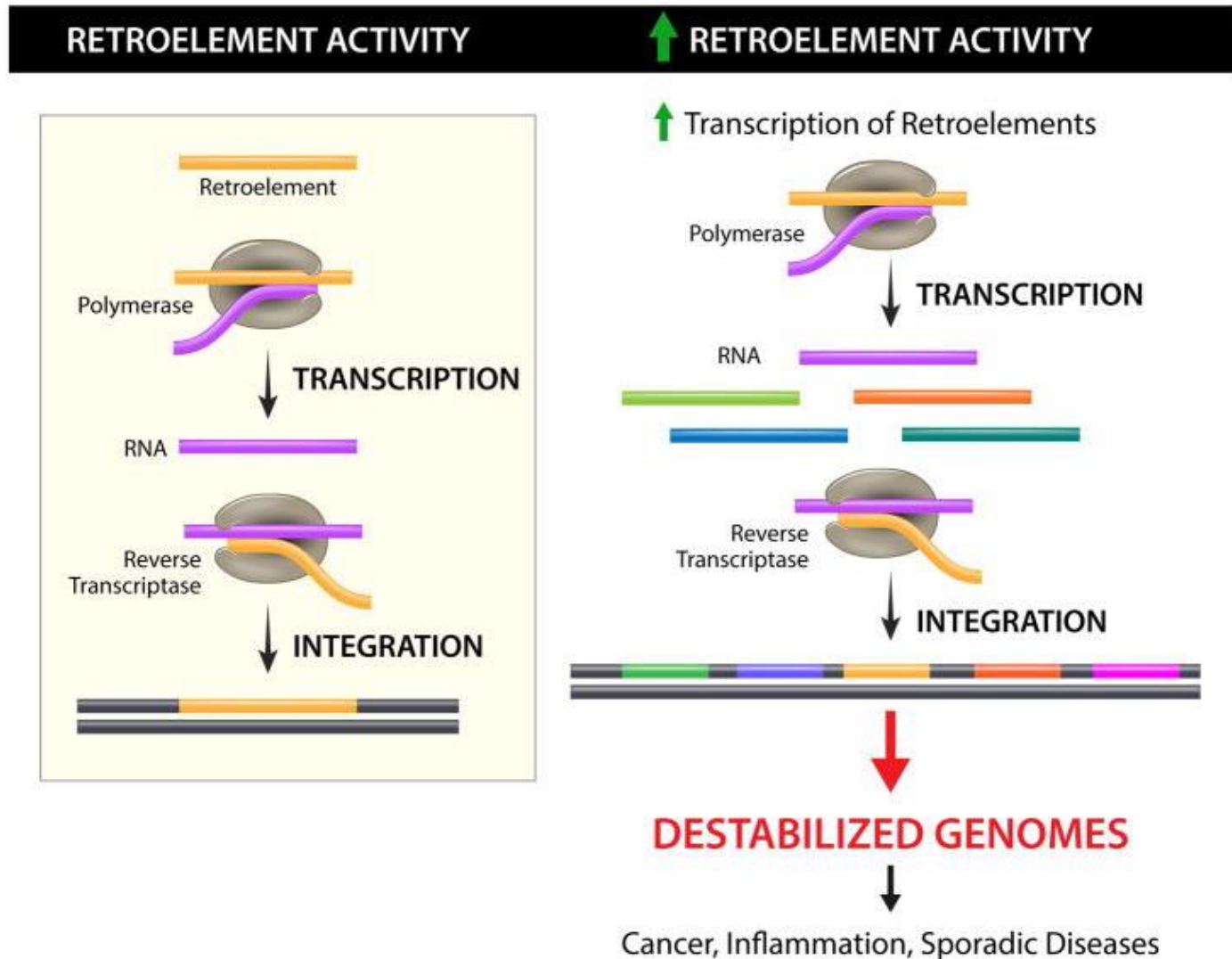


Transposable elements are abundant in the human genome, and great strides have been made in pinpointing variations in these repetitive sequences using whole-genome sequencing. **Now, the focus is shifting to understanding their expression and regulation**, and the functional consequences of their insertion and retention in the genome over time. Whereas transposable element insertions have been known to cause human genetic disease since the 1980s, the scope of their contributions to heritable phenotypes is now starting to be uncovered. Here, we review the many ways human retrotransposons contribute to genome function, **their dysregulation in diseases including cancer** and how they affect genetic disease.

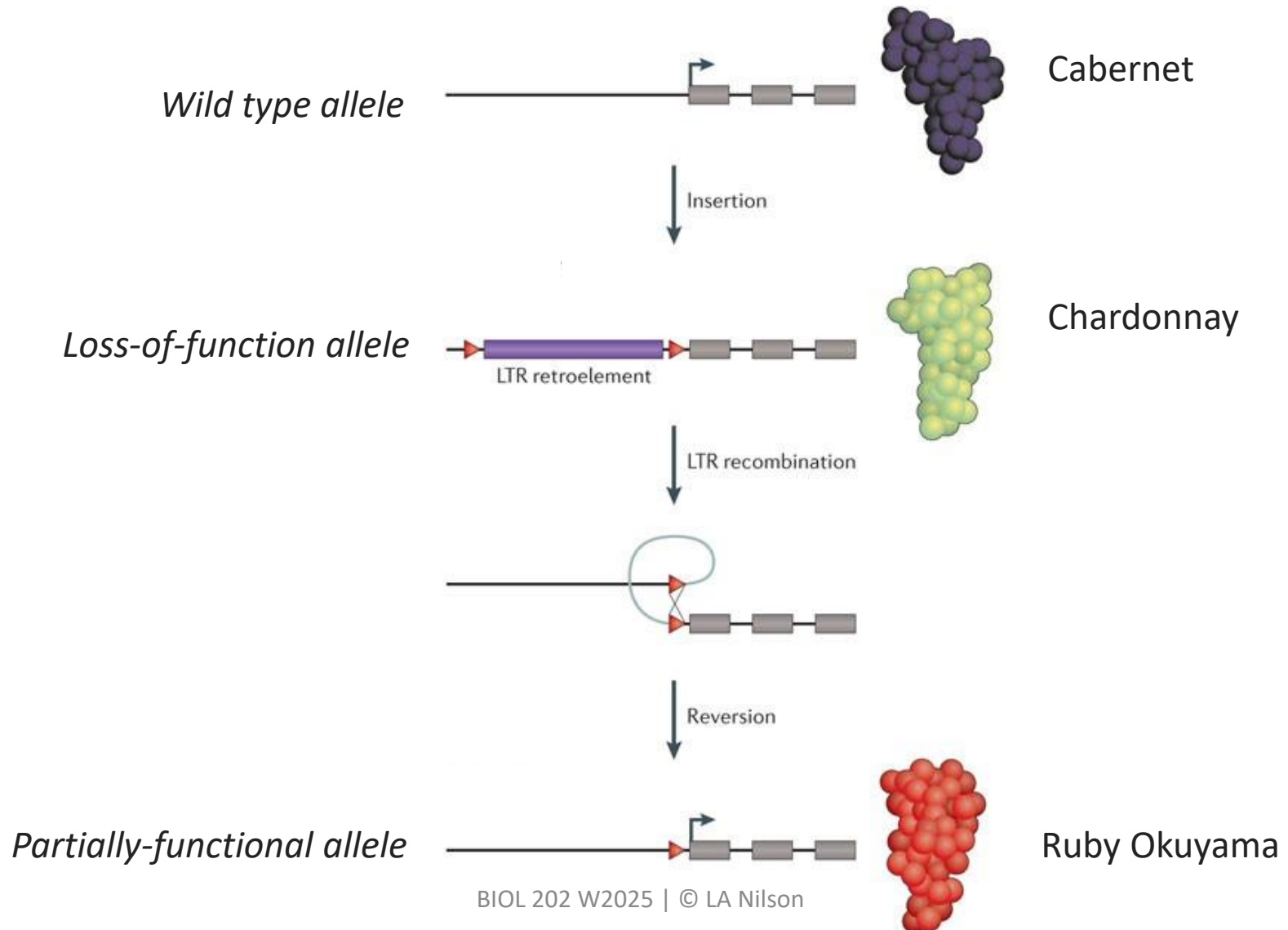
The Dynamic Genome

- Discovery of transposable elements through study of “unstable” alleles
- Impact of transposable elements on the genome
- Genomic surveillance mechanisms that limit transposon mobility

Transposable element mobility can destabilize genomes



Transposable elements as a potential a source of major phenotypic novelty and adaptive change



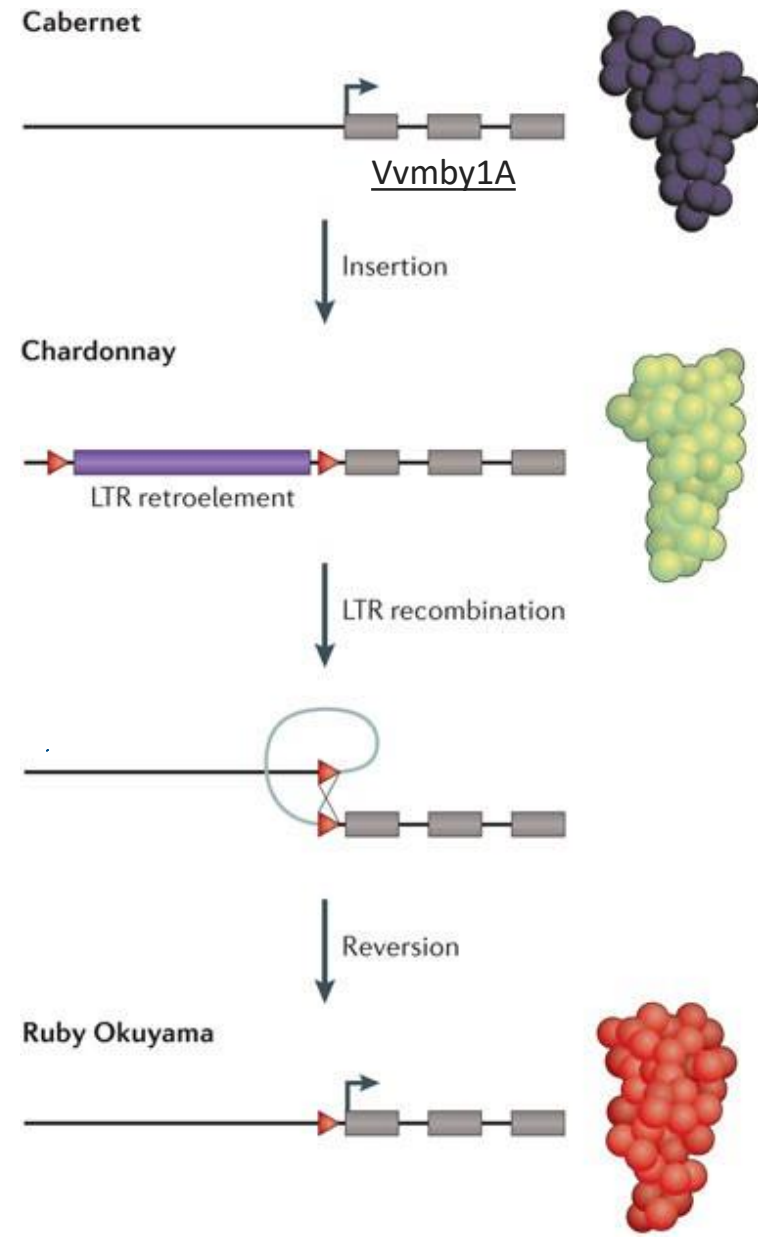
Transposable elements as a source of phenotypic novelty

The *Vvmyb1A* gene is required for production of purple pigment in Cabernet grapes.

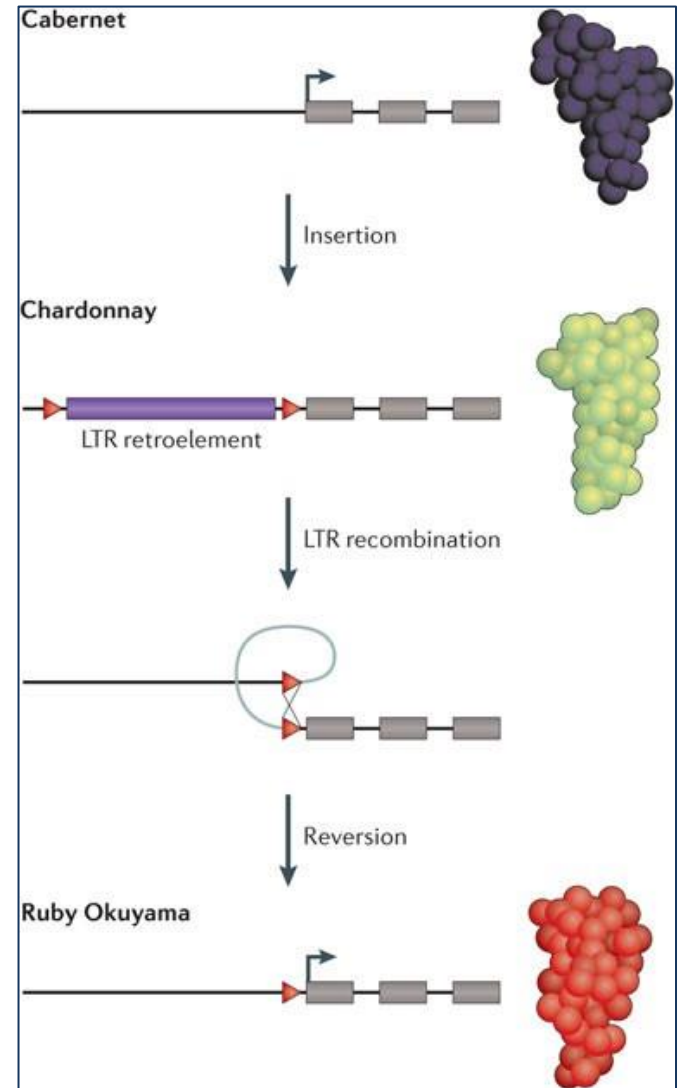
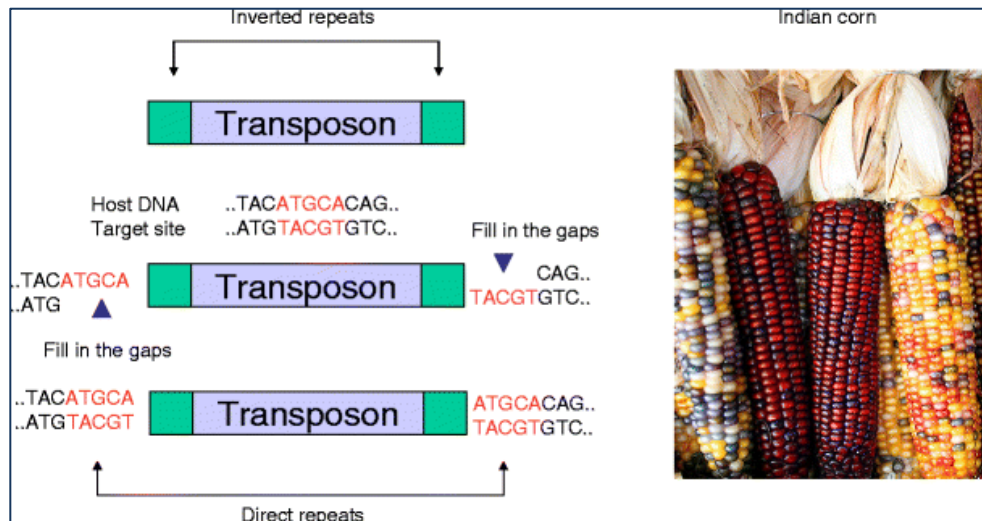
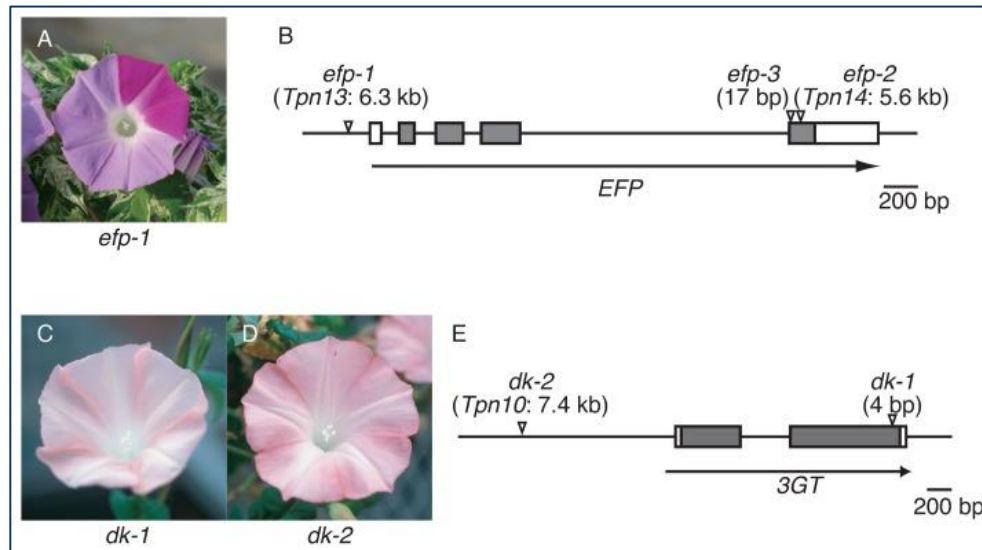
An initial insertion of a *Gret1* long terminal repeat (LTR) retrotransposon resulted in a loss-of-function allele of the *Vvmyb1A* gene, leading to a loss of colour in the fruit of the Chardonnay variety.

A subsequent rearrangement in *Gret1* results in revertant, coloured grapes in varieties such as Ruby Okuyama.

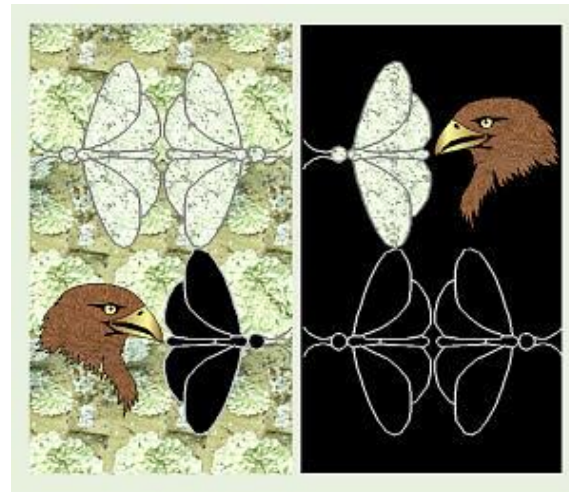
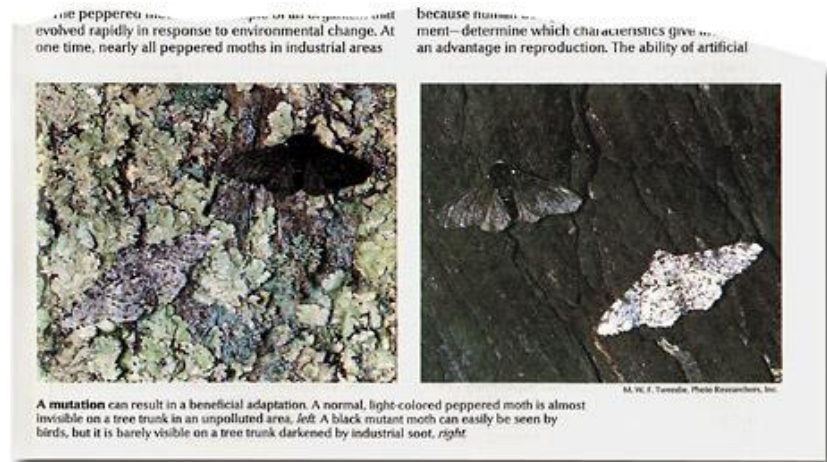
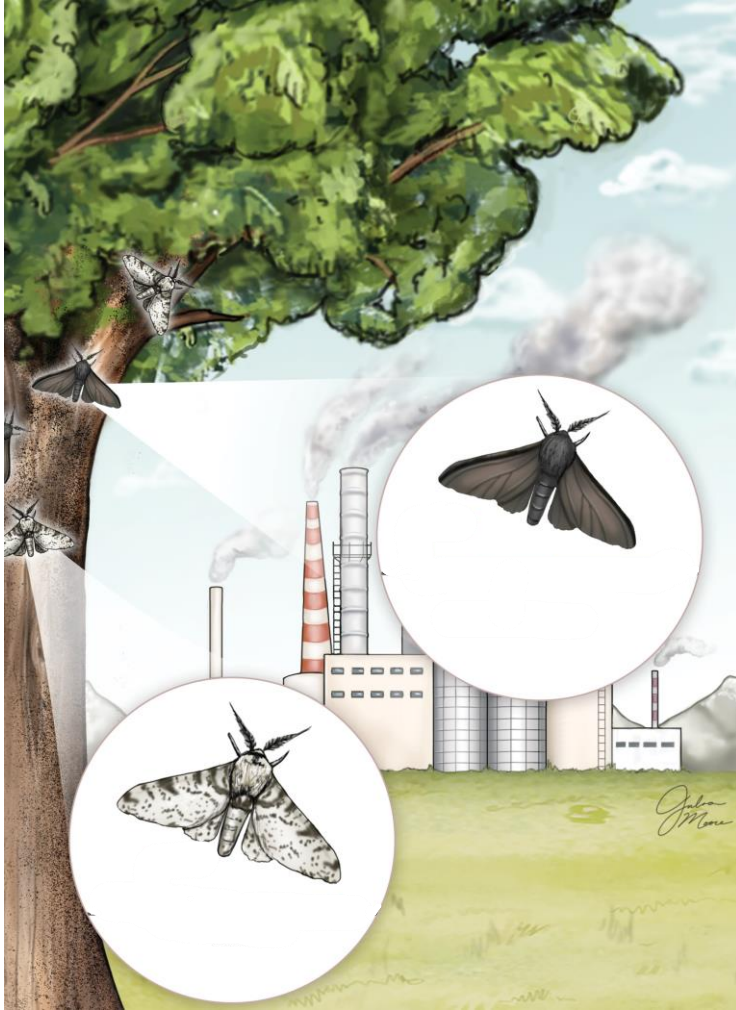
Exons are depicted as grey boxes. The LTRs flanking Gret1 just upstream of the Vvmyb1A gene are depicted as red triangles.



Transposable elements as a potential source of major phenotypic novelty

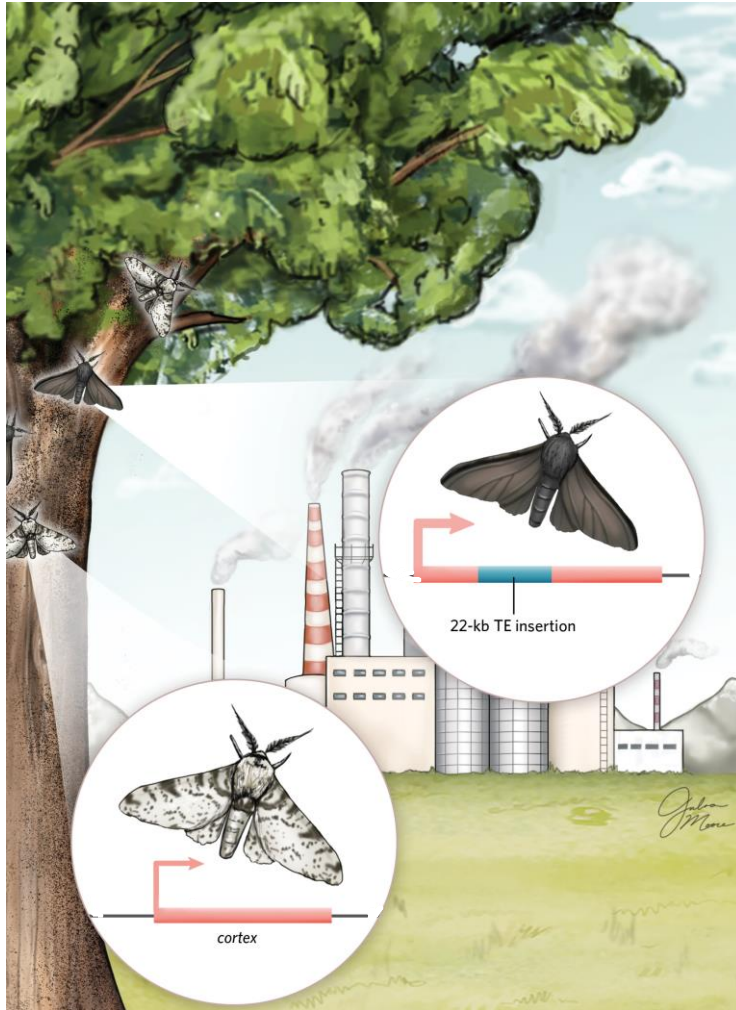


Transposable elements as a potential source of major phenotypic novelty and adaptive change



[Peppered Moths | Natural Selection Game \(asu.edu\)](http://asu.edu)

Transposable elements as a potential a source of major phenotypic novelty and adaptive change



(excerpt from scientific article in the journal Nature, in 2016 – see link below)

“Here we show that the mutation event giving rise to industrial melanism in Britain was the insertion of a large, tandemly repeated, transposable element into the first intron of the gene cortex. Statistical inference based on the distribution of recombined carbonaria haplotypes indicates that this transposition event occurred around 1819, consistent with the historical record.”

“The discovery that the mutation itself is a transposable element will stimulate further debate about the importance of 'jumping genes' as a source of major phenotypic novelty.”

Genome Surveillance

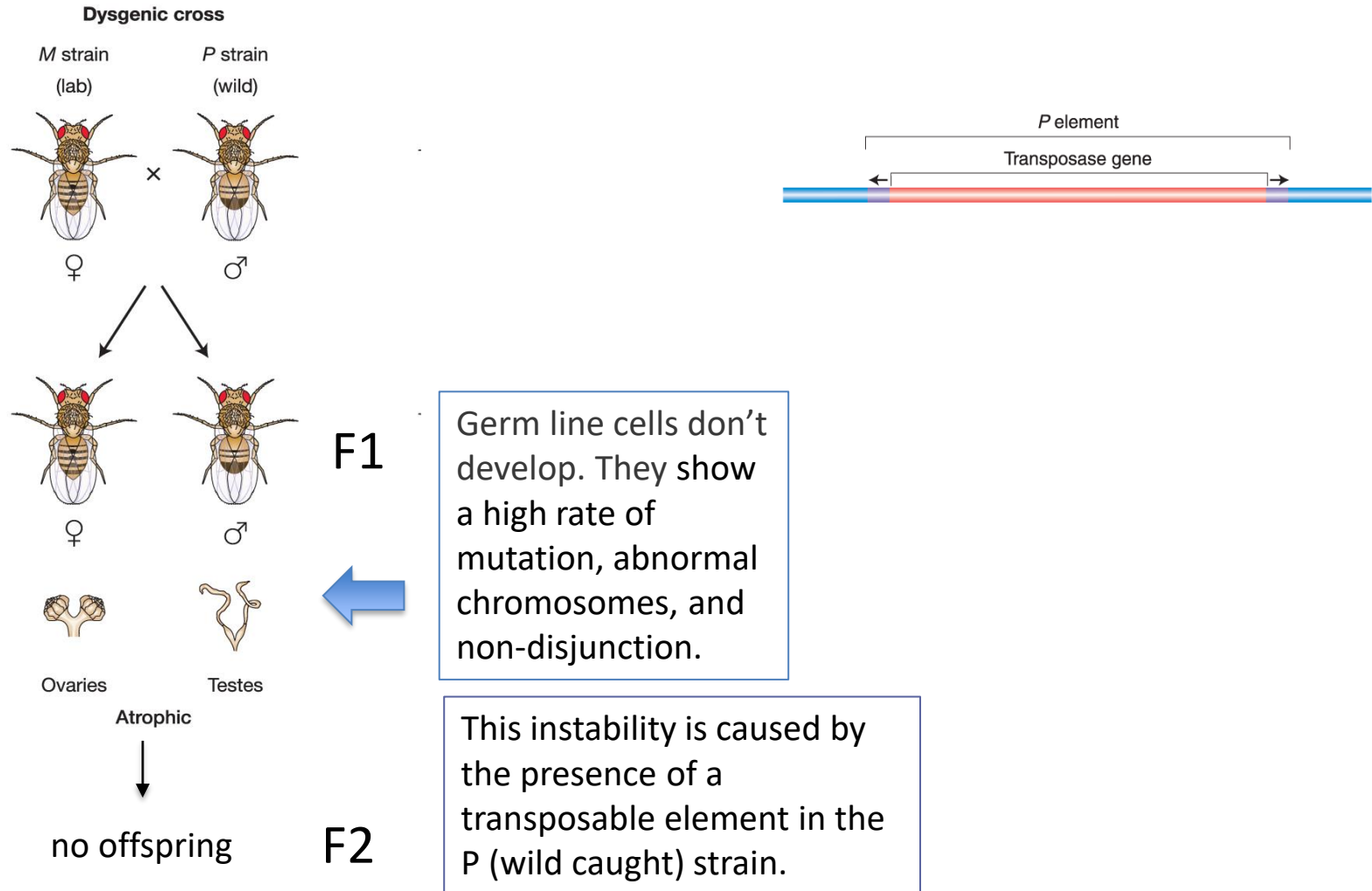
Limiting the impact of transposable element mobility

Two general points to keep in mind:

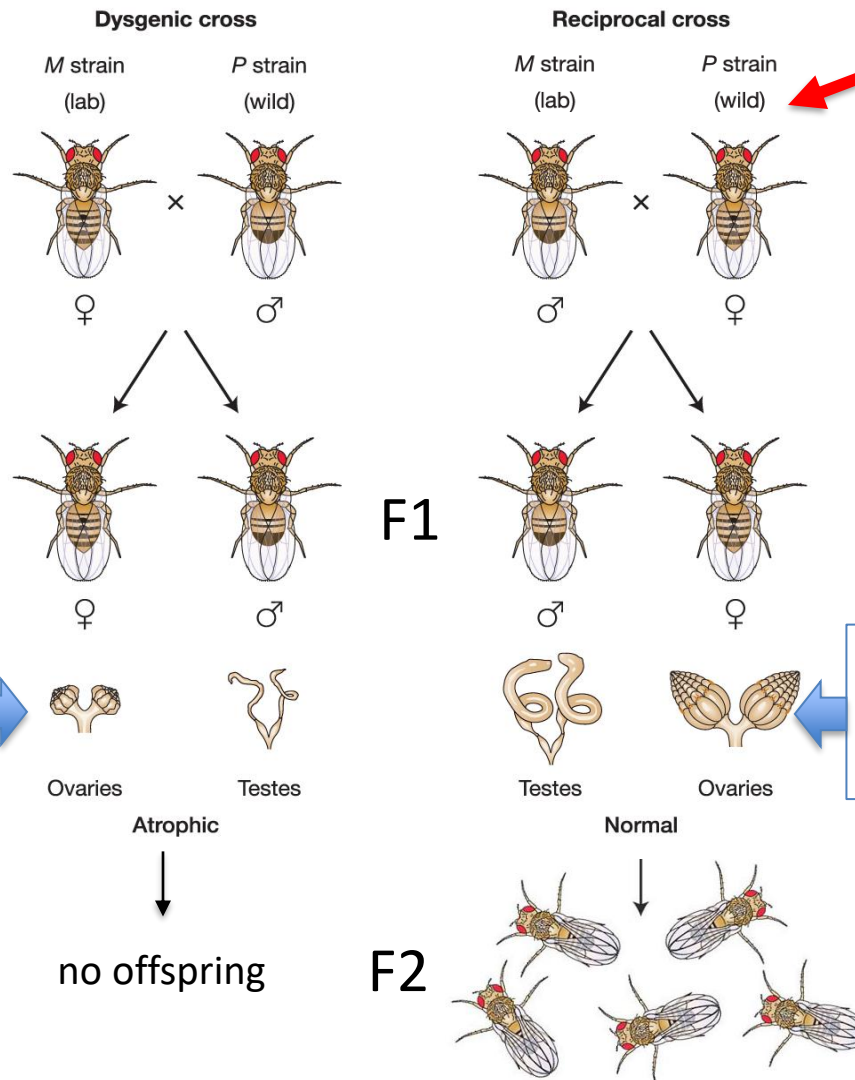
- Transposable element mobility leads to mutations and genome instability.
- Cells have mechanisms to repress transposable element mobility and protect genome integrity.

Also: understand the experiments, results, and interpretations

The Drosophila P-element (a type 1 transposon) was discovered by mating lab strains with wild-caught strains



Drosophila P-elements were discovered by mating lab strains with wild-caught strains



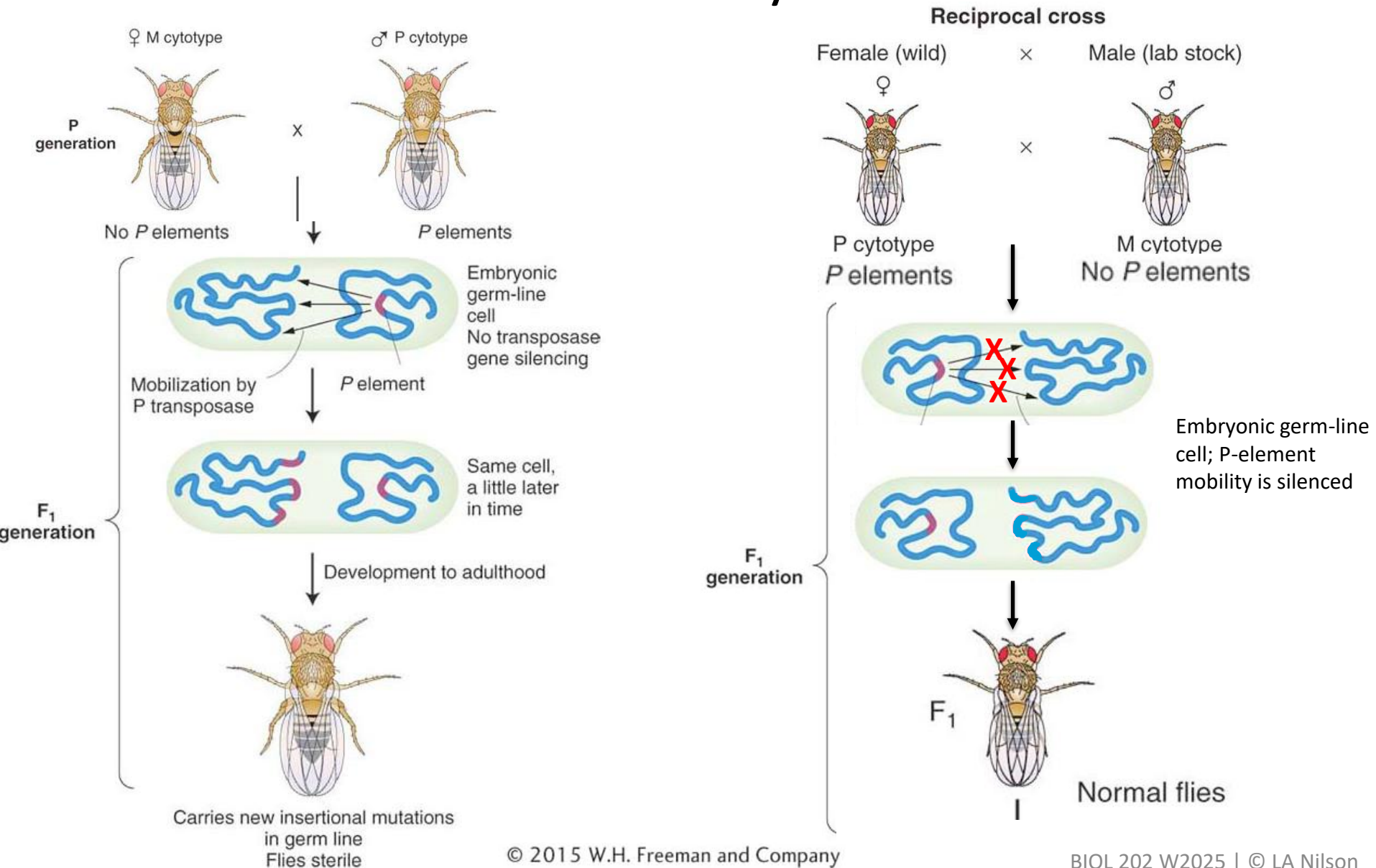
warning:: there is a mistake in these labels in the version of the picture in the textbook. This version here is correct.

If the P strain contains transposable elements, why does this cross yield normal F2 progeny?

Observation: In this cross, the genome of the germ cells remains intact despite the presence of P elements.

Interpretation: A mechanism must exist that preserves genome integrity, perhaps by blocking P-element transposon mobility

Mechanisms exist that repress transposable element mobility



Cells have mechanisms that repress transposable element mobility.

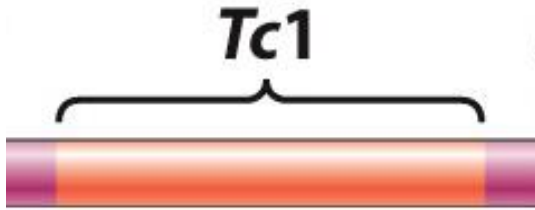
What are they and how do they work?

How does a cell silence its transposons?

What is the molecular mechanism of transposon silencing?

How do we figure this out?

C. elegans also have transposable elements, plus mechanisms for repressing transposable element mobility



C. elegans Tc1 element

an autonomous DNA transposon, analogous to the maize Ac element, and the *Drosophila* P element.

32 Tc1 element copies in the *C. elegans* genome



Observation: Tc1 elements transpose in somatic cells but not germline cells.

Interpretation: The transposase gene in each Tc1 copy is silenced in germline cells.

Hypothesis: The germline cells contain some kind of “silencing machinery” that represses Tc1 mobility. **The components of this “machinery” are encoded by genes.**

How do we identify the genes that cells use to repress transposable element mobility?

Rationale:

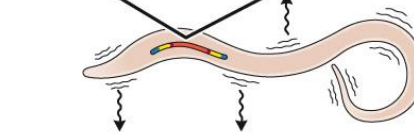
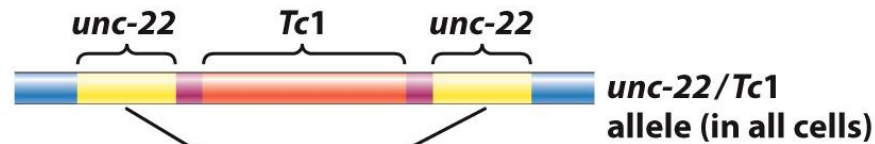
- If we knock out a gene involved in repressing transposable element mobility*, then transposable elements will be able to move.
- Conversely, if we find evidence that a mutant strain shows evidence of transposable element mobility, then we can infer that the mutated gene is required to repress transposable element mobility.
- Figuring out which gene is mutated in such a strain will identify a part of the cellular machinery that represses transposable element mobility.

*Q: If we don't know how the process works, how do we know which genes to knock out?

If transposon mobility is no longer repressed, Tc1 can jump out of the *unc-22* gene, restoring *unc-22* function

Tc1 insertion in the *unc-22* gene disrupts *unc-22* function

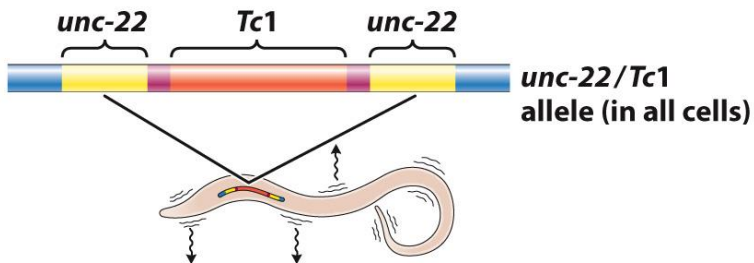
Approach: Do a "Genetic Screen" to identify mutations that allow Tc1 to mobilize. We then infer that the knocked-out genes are involved in **silencing** of Tc1 mobility.



Twitching/uncoordinated worm

Derivative mutant strains

If Tc1 does not mobilize
("jump")



Twitching/uncoordinated worm

If Tc1 mobilizes
("jumps")



Smooth-gliding worm

F1

Genetic screen: mutate genes at random, then select those in which *unc-22* function is restored.

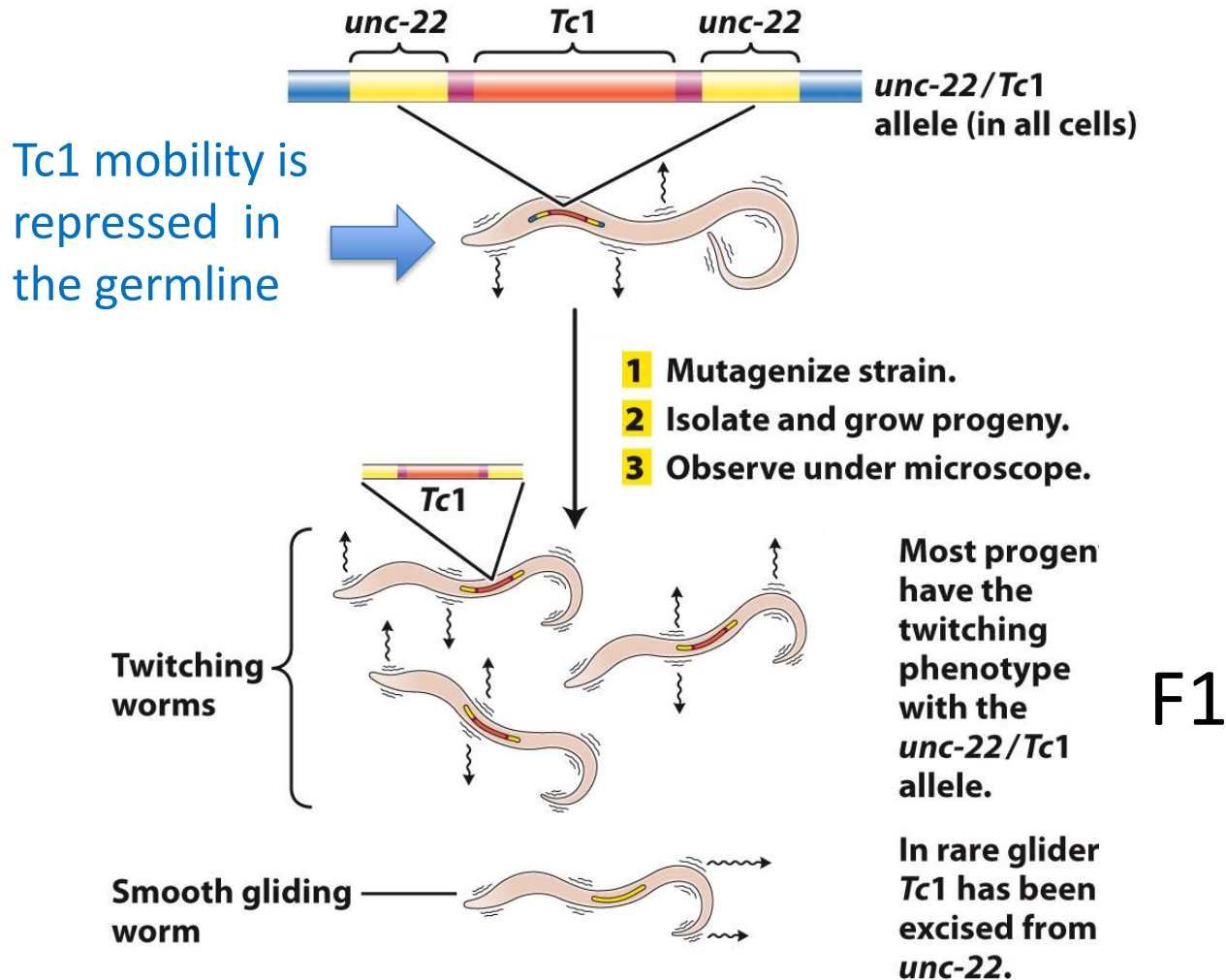


Figure 15-25
Introduction to Genetic Analysis, Eleventh Edition
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How do we identify the genes that cells use to repress transposable element mobility?

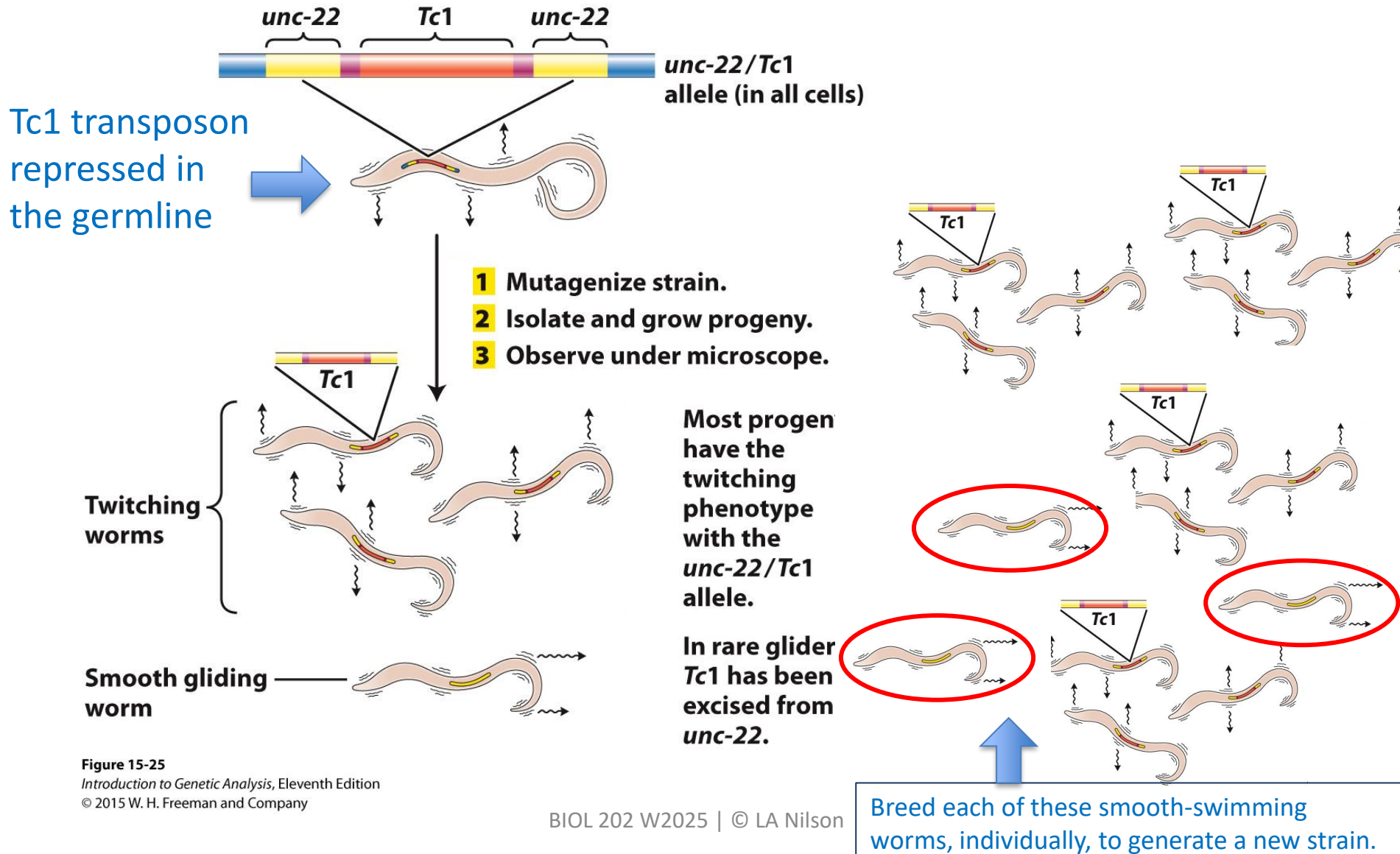


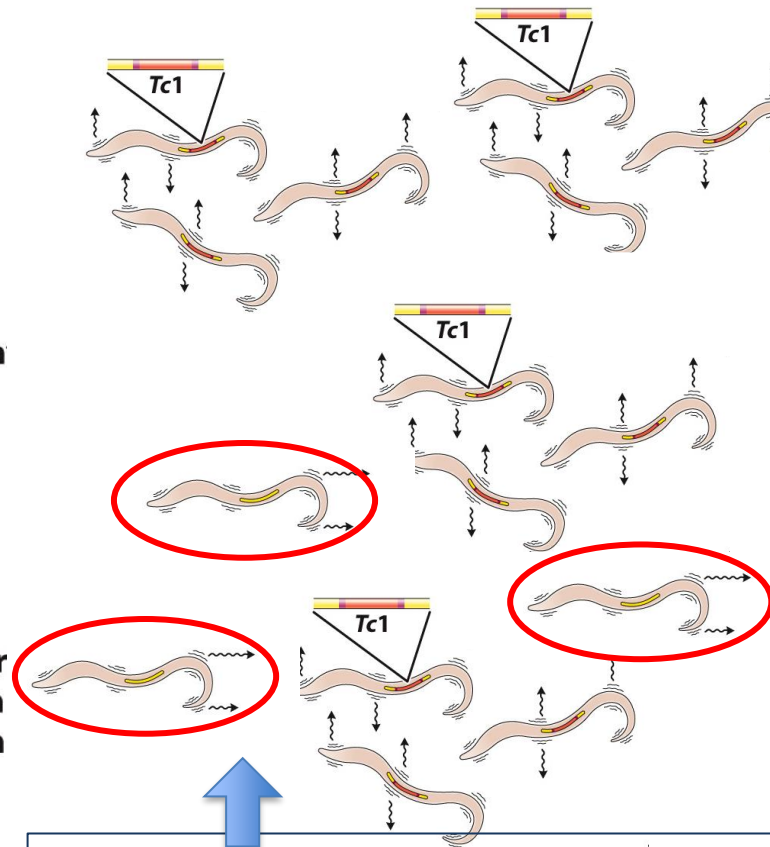
Figure 15-25
Introduction to Genetic Analysis, Eleventh Edition
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Think Break

What happened to these worms?

Most progen have the twitching phenotype with the *unc-22/Tc1* allele.

In rare glider *Tc1* has been excised from *unc-22*.



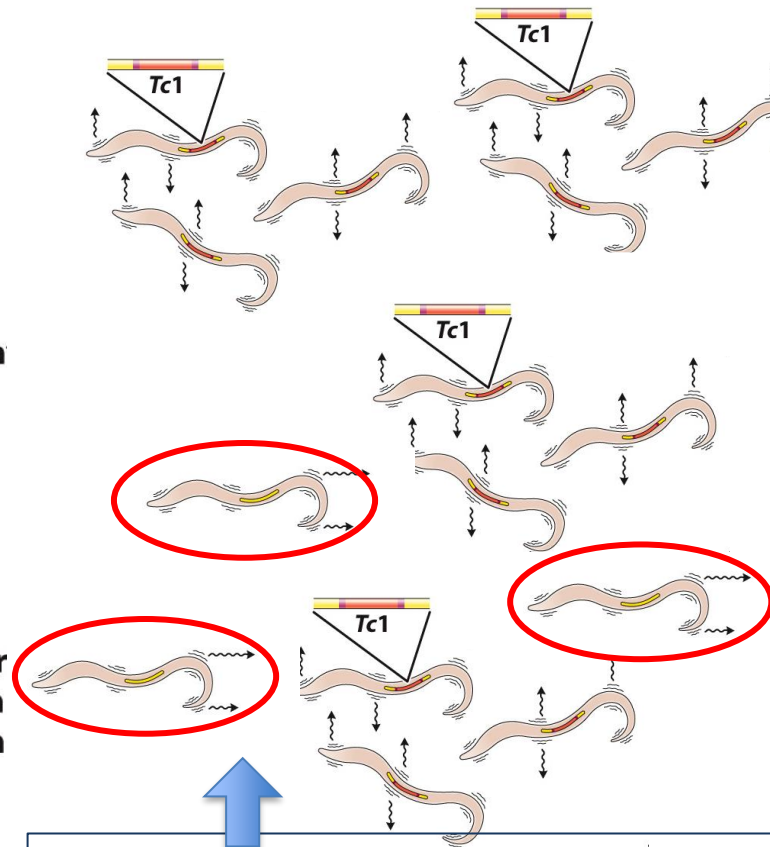
What happened to these worms?

Is the same gene knocked out
each worm?

How would you tell?

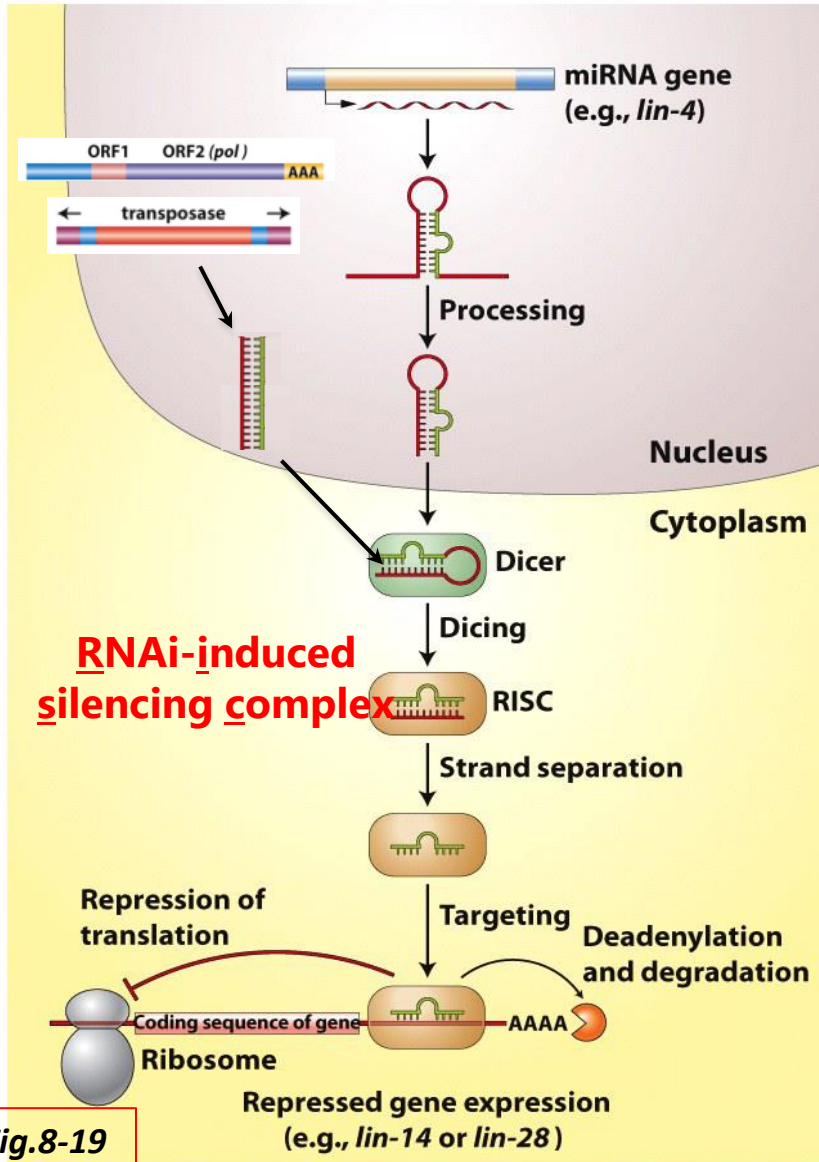
Most progen
have the
twitching
phenotype
with the
unc-22/Tc1
allele.

In rare glider
Tc1 has been
excised from
unc-22.



Screens identify 25 genes that block Tc1 mobility.

Many of these genes are involved in the RNAi silencing pathway.



Based on short (~21-25 nt) RNAs derived from double-stranded RNA precursors

Processing involves many steps, including cleavage by Dicer

Once fully processed, miRNAs bind to RISC complex

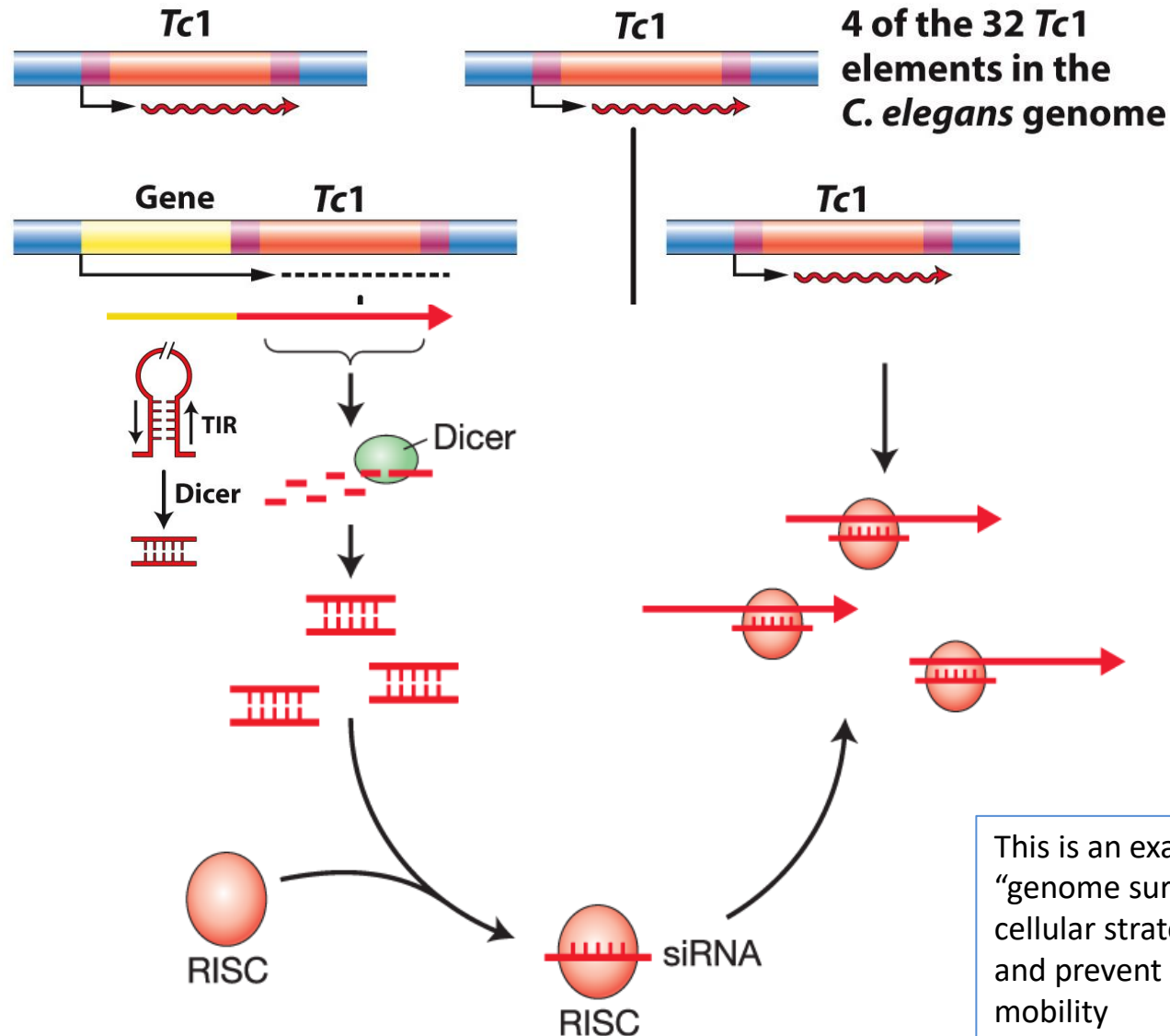
miRNA/RISC complex recognizes mRNA targets through sequence complementarity, and either promotes their degradation or represses their translation

microRNA (**miRNA**): endogenous gene

siRNA: foreign gene (transgene, virus, TE)

Fig.8-19

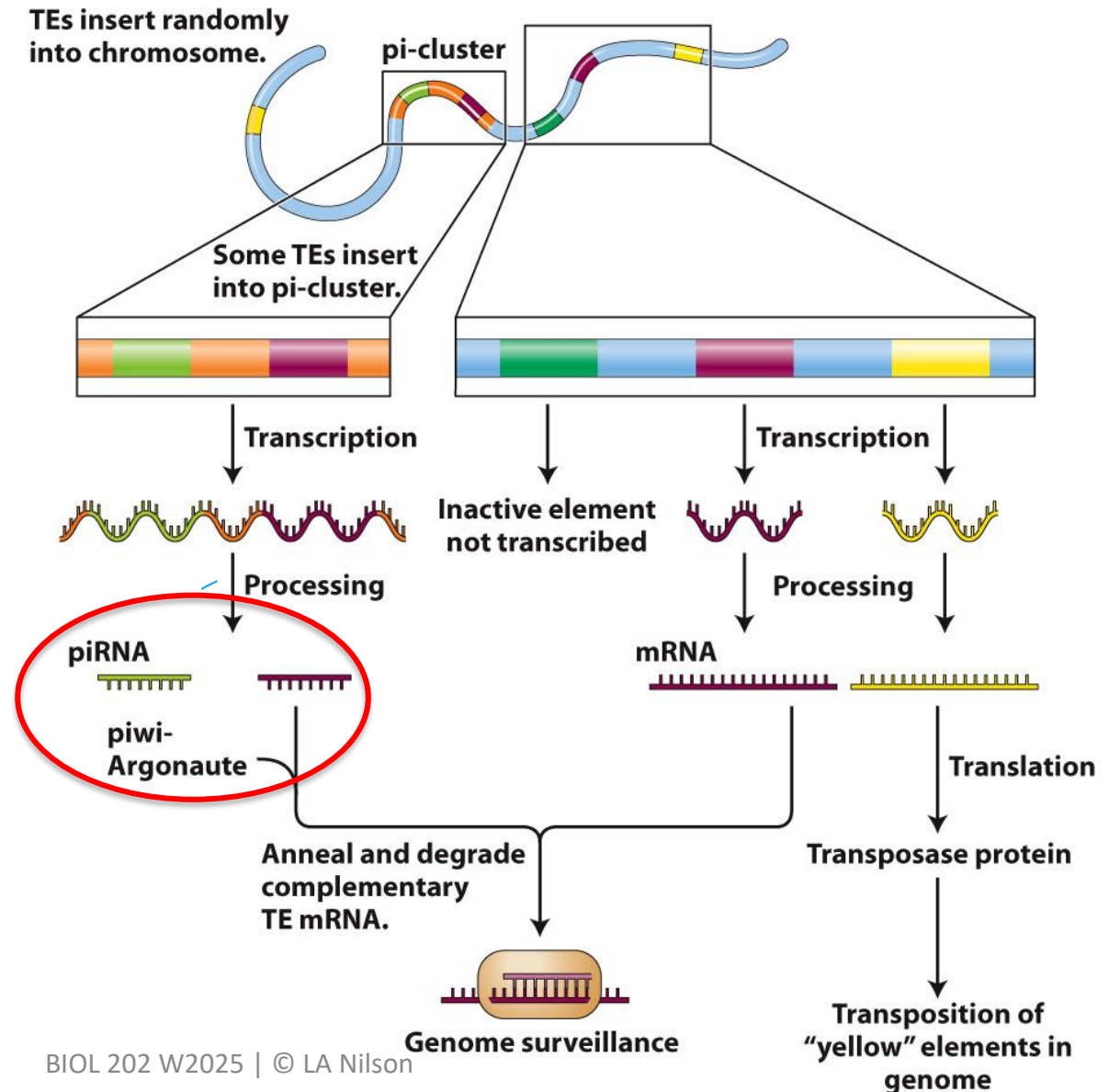
A single Tc1 can repress all Tc1 mobility in a cell



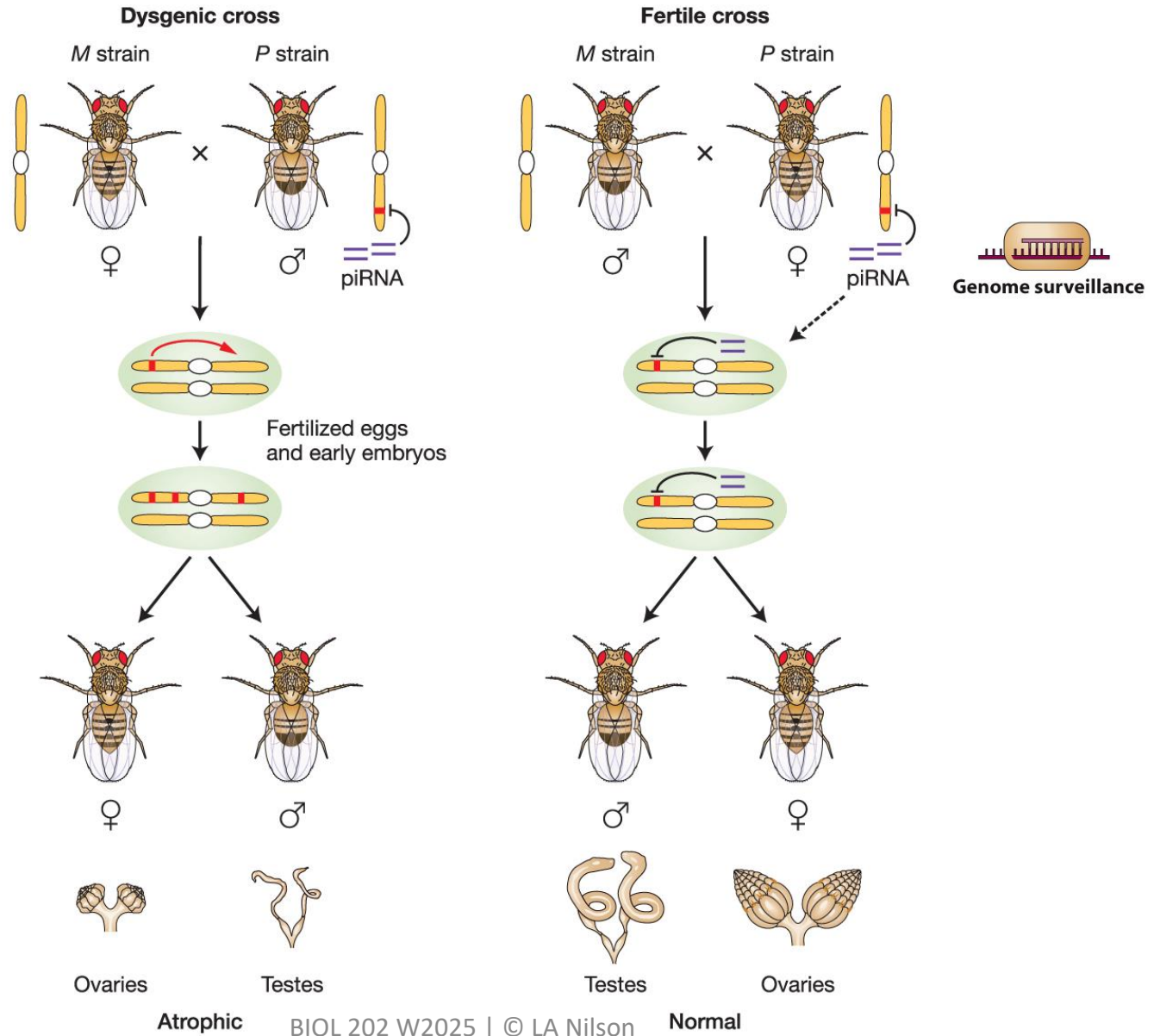
piRNAs repress transposon mobility in the germ lines of *Drosophila* and other animals

Like siRNAs, piRNAs are short single-stranded RNAs (26–30 nt in mammals) that interact with a protein complex that destroys complementary target mRNAs.

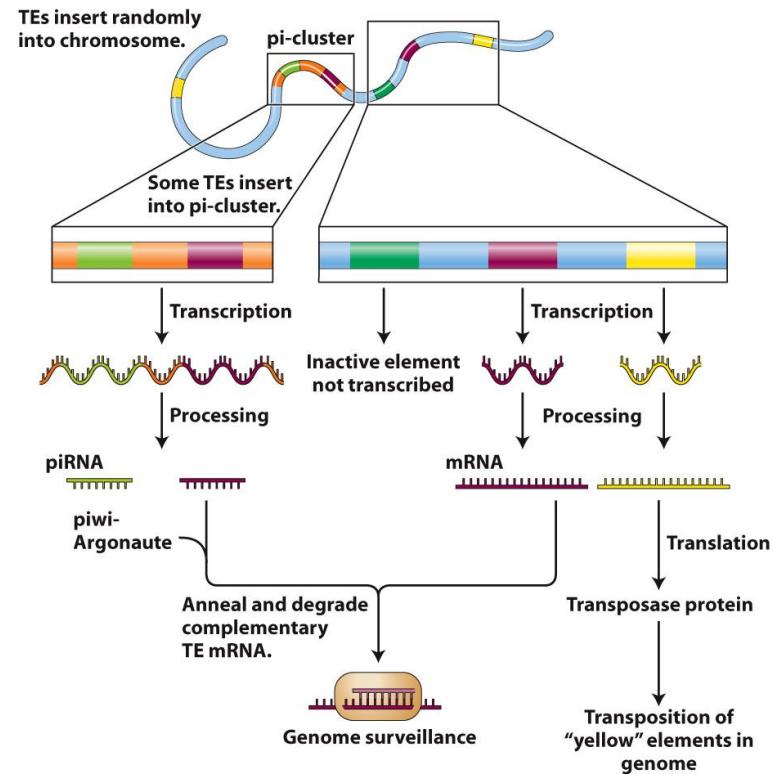
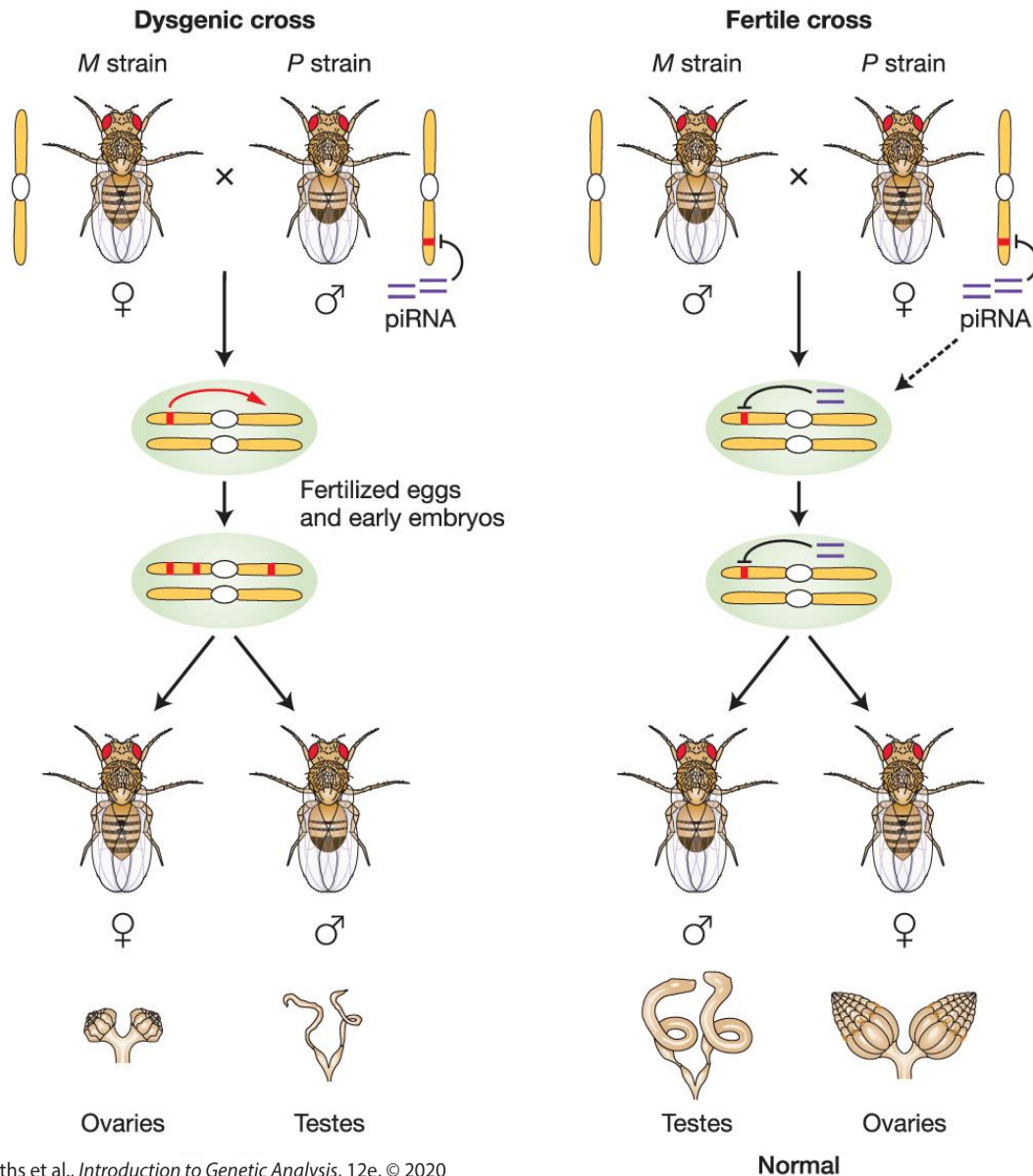
Unlike siRNAs, piRNAs do not start out as double-stranded RNA, but instead as parts of long mRNAs from a specific genome region called a pi-cluster.



“Hybrid dysgenesis” explained: piRNAs are present in P strain but not M strain



“Hybrid dysgenesis” explained: piRNAs are present in P strain but not M strain

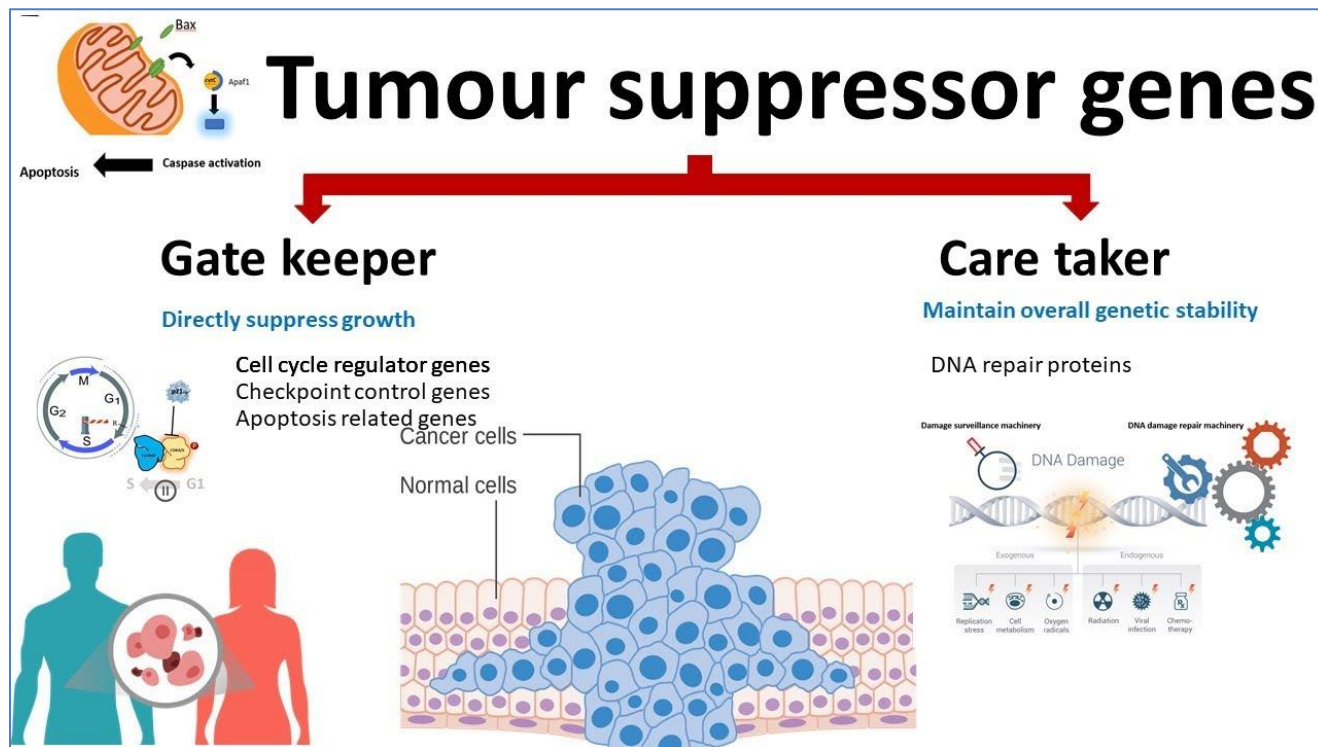


Repression of transposable element mobility in humans may suppress tumor formation

The p53 gene is one of the most studied mutations in human cancer.

Mutations in p53 are found in most tumor types and contribute to the complex network of molecular events leading to tumor formation.

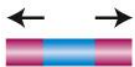
Hypothesis: p53 plays a role in repressing transposable element mobility.

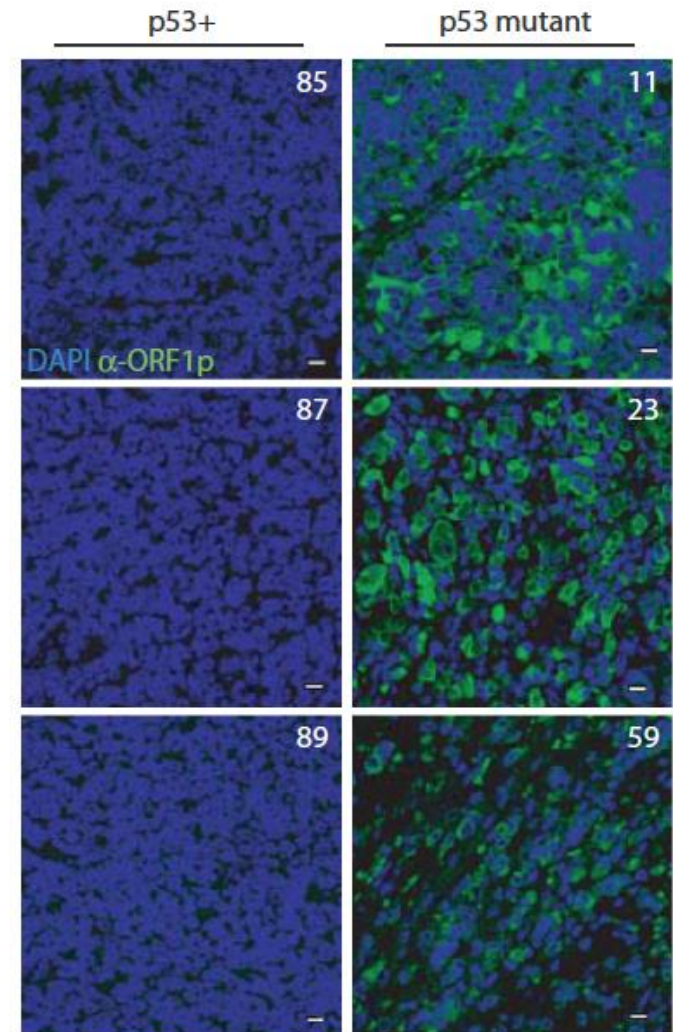


Wilms (kidney) tumors mutant for p53 show elevated human LINE1 ORF1p expression

blue = DNA

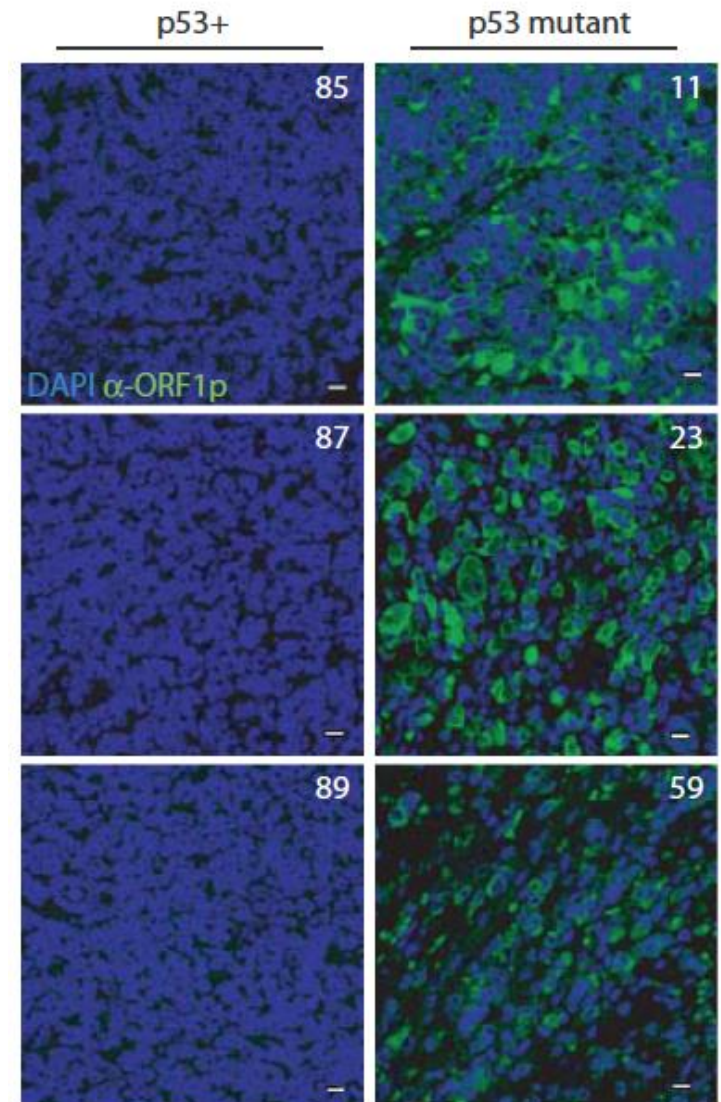
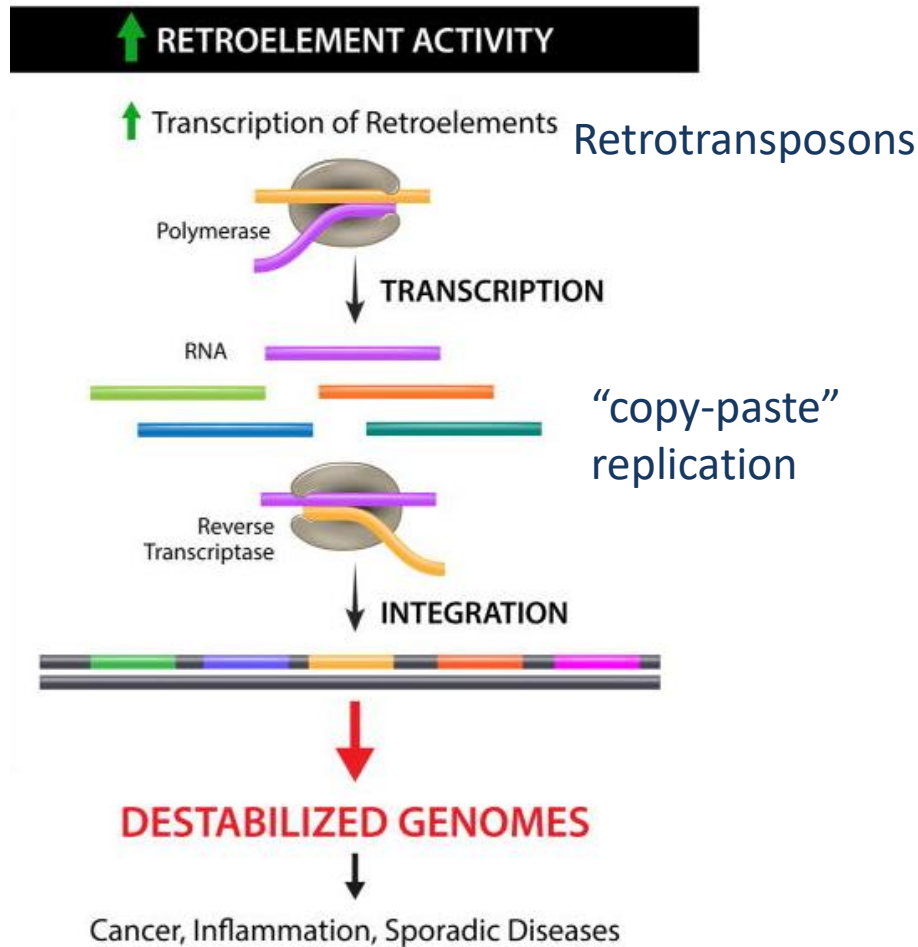
green = human LINE-1 ORF1p expression

Element	Transposition	Structure
LINES	Autonomous	
SINEs	Nonautonomous	
DNA transposons	Autonomous	
	Nonautonomous	

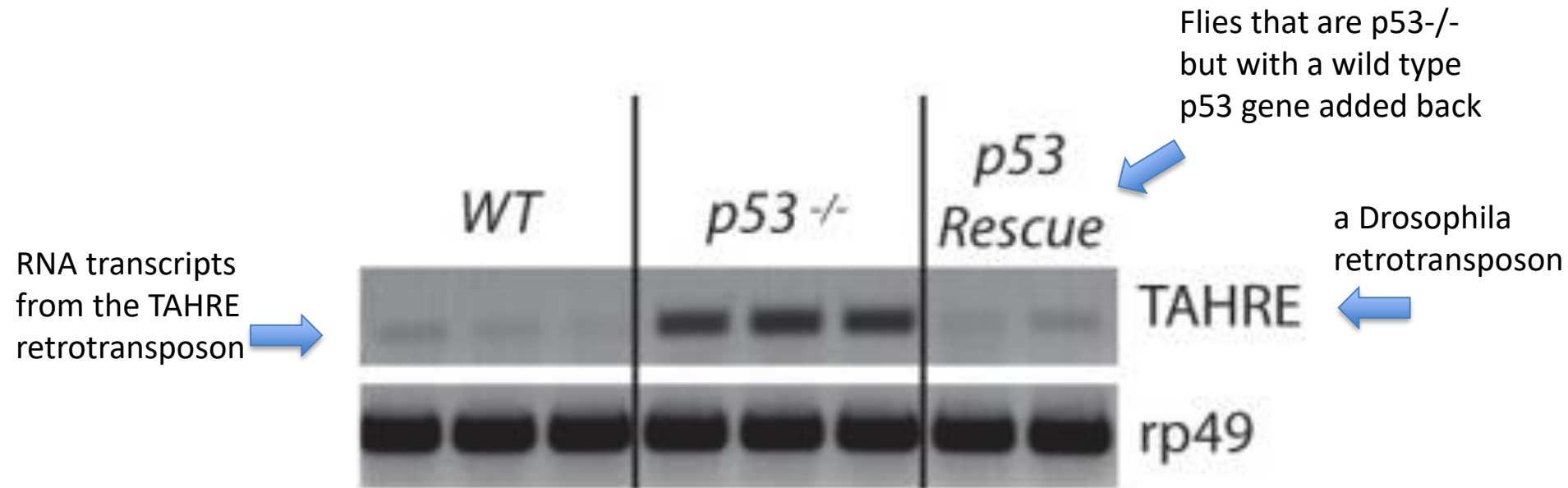


Deregulated retroelements stratify with p53 mutations in Wilms tumors. Compared with Wilms tumors that are wild type for p53 (*left* panels: 85, 87, and 89), Wilms tumors that are mutant for p53 (*right* panels: 11, 23, and 59) show dramatically elevated human LINE-1 ORF1p expression (α -ORF1p [green]; counterstained to visualize DNA [blue])

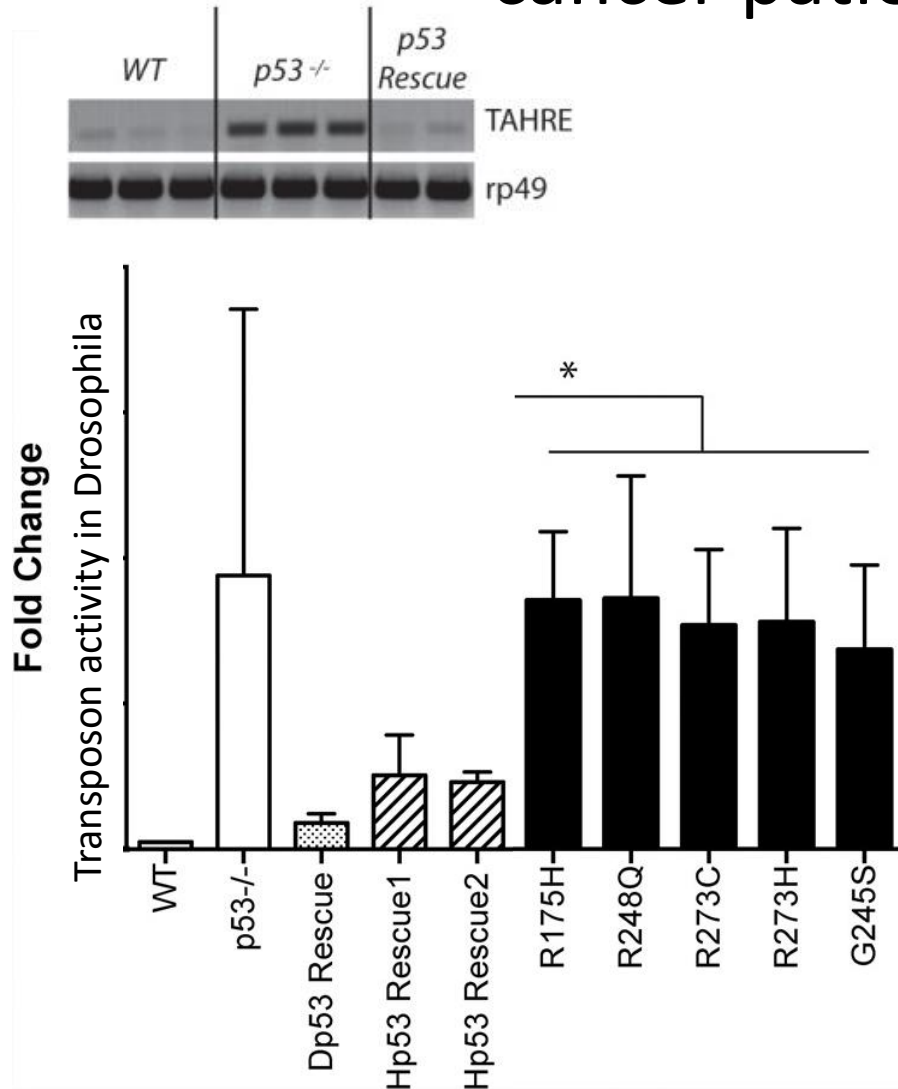
Hypothesis: Loss of p53 in kidney cells leads to deregulation of retrotransposon activity and to tumorigenesis.



The p53 protein is required for silencing transposon mobility in *Drosophila*



Human p53 corrects dysregulated transposon activity in p53^{-/-} flies, but p53 variants commonly seen in cancer patients do not.

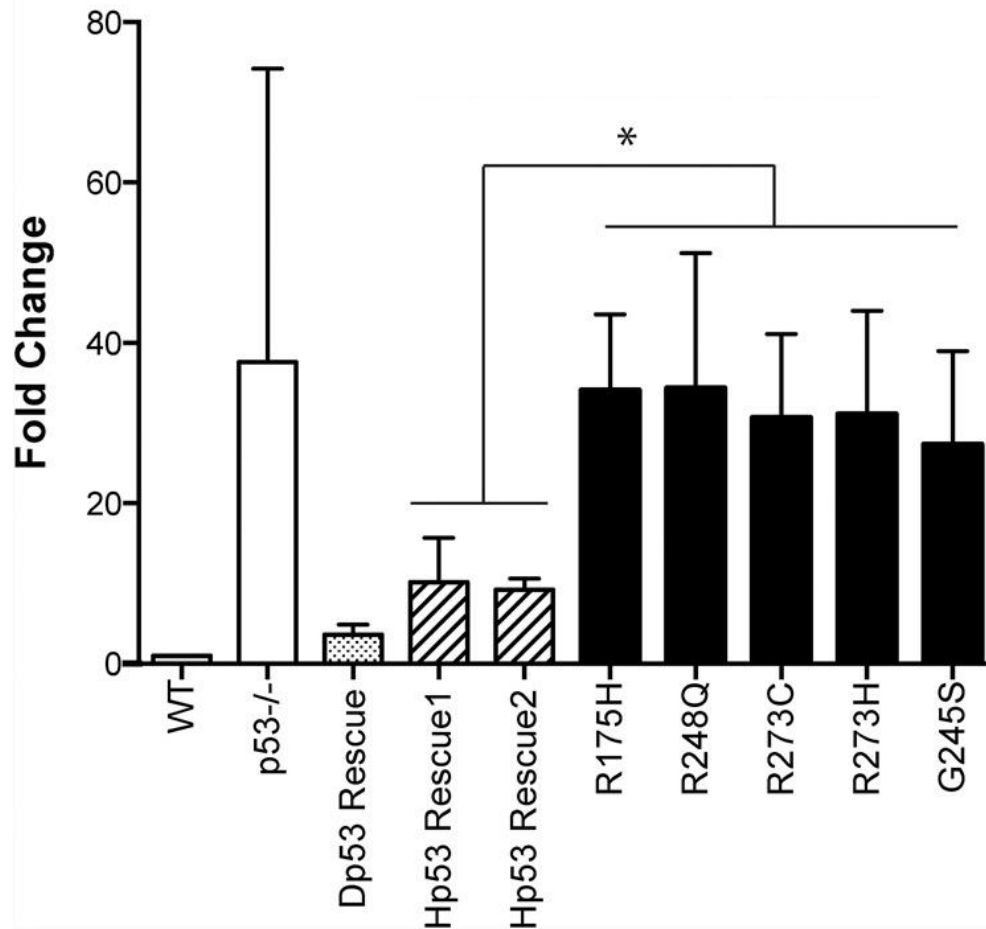


- WT : wild type fly
- p53^{-/-} : P53 mutant fly
- Dp53 Rescue: p53 mutant fly + transgenic wild type Drosophila p53
- Hp53 Rescue1: p53 mutant fly + transgenic wild type human p53
- Hp53 Rescue2: p53 mutant fly + transgenic wild type human p53 (replicate)
- Black bars: p53 mutant fly + a transgenic human mutant p53 allele from a cancer patient

Think Break

Data Interpretation

Which strain has the most p53 activity?



The p53 protein promotes genome stability by silencing transposon mobility

